

M. SHAHARYAR

Founding Engineer | Real-Time Systems, Applied AI & Product Architecture | Remote Worldwide | shaharyar321321@gmail.com | shaharyar.dev | [Github](https://github.com/shaharyar) | [LinkedIn](https://www.linkedin.com/in/shaharyar)

SUMMARY

Founding engineer specializing in real-time systems, AI-native products, and 0-to-1 SaaS development. Built and operated collaborative platforms, semantic-search infrastructure, workflow systems, and developer tooling serving 1,000+ beta users. Strongest in TypeScript and Rust systems where ambiguous product requirements need to become reliable shipped software.

EXPERIENCE

NOVUS ATLAS (NOVA) — Founder & Lead Engineer

Jun 2024 - Present novusatlas.org | codex.novusatlas.org

- Sole engineer responsible for architecture, implementation, infrastructure, deployment, and technical documentation across a multi-surface fiction platform serving **1,000+ beta users**.
- Engineered real-time collaborative editing with Operational Transformation (OT), WebSockets, presence awareness, autosave, optimistic synchronization, and offline-resilient recovery; sustained **500+ concurrent collaborators at <80 ms p95 synchronization latency**.
- Designed Rust-powered semantic-search and content-analysis pipelines using vector embeddings, Qdrant, PostgreSQL, and LLM-assisted processing; **indexed 5M+ searchable content chunks** with **<150 ms p95 retrieval latency**.
- Created Git-style branch-based version control for fiction and shipped platform infrastructure across Next.js, Hono/oRPC, Redis, BullMQ, Cloudflare R2, Socket.IO, Rust/N-API, Railway, and Vercel.

KICKLAYER — Founder & Lead Engineer

2026 - Present kicklayer.com

- Built a multi-tenant workflow platform that orchestrates client onboarding, asset collection, credential exchange, compliance tracking, and AI-assisted project preparation for agencies.
- Designed schema-driven onboarding workflows with conditional logic, progress tracking, file validation, version history, automated reminders, authenticated exports, and field-level audit trails.
- Built secure client infrastructure featuring passwordless authentication, AES-256-GCM credential storage, secret detection, access controls, token revocation, and activity tracking.
- Integrated AI-assisted asset verification, onboarding analysis, project brief generation, delay-risk detection, subscription enforcement, usage metering, and billing workflows.
- Reduced onboarding **follow-up workload by 61%** and shortened **asset-collection timelines by 54%**.

SELECTED ENGINEERING PROJECTS

OPENOT — Open-Source Operational Transformation Framework - open-ot.shaharyar.dev

- Built a framework-agnostic OT engine with state-machine transforms, pluggable WebSocket and SSE transports, storage adapters, and React/Lexical integration kits.
- Validated correctness through **10,000+ transformation test cases** and benchmarked throughput exceeding **150,000 operations per second**.

THE ARCHITECT ENGINE — AI-Native Narrative Platform - 2026 architect.novusatlas.org

- Built an AI-native narrative platform featuring adaptive story interviews, structured generation, branching scenario chains, persistent chronicles, and stateful orchestration.
- Designed server-side prompt architecture, typed JSON contracts, retry pipelines, world-scoped configuration, and model-routing strategies.
- **Reduced perceived generation latency to <1.2 seconds** while **lowering inference costs per session by 72%**.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Rust, Python, SQL

Frameworks & Infrastructure: React, Next.js, Node.js, Hono, tRPC, oRPC, PostgreSQL, Redis, Qdrant, Cloudflare R2, Docker, Railway, Vercel

Systems & AI: Operational Transformation, distributed synchronization, WebSockets, SSE, event-driven systems, LLM orchestration, structured generation, model routing, vector embeddings, semantic search, NLP pipelines, AES-256-GCM

TECHNICAL WRITING

Architecture documentation and engineering deep-dives: codex.novusatlas.org